

EE619 Term Project

School of Electrical Engineering, KAIST

Due by 10:00 PM on June 7, 2026

Overview

Implement a Deep Q-Network (DQN) agent for `MountainCar-v0`. MountainCar has a continuous two-dimensional state, [position, velocity], and three discrete actions: accelerate left, do nothing, and accelerate right. The goal is to reach the flag on the right hill. Because the engine is weak, the agent usually needs to build momentum by moving back and forth before reaching the goal.

The main purpose of this project is not only to make a DQN run, but to understand how exploration affects learning in a delayed-reward control problem.

Notebook Tasks

Complete the provided notebook. The implementation is intentionally split into several components.

You must compare two epsilon-greedy exploration schedules:

- **Less exploration:** epsilon decreases quickly, final epsilon is small
- **More exploration:** epsilon decreases slowly, final epsilon is large

For each schedule, run three random seeds:

$$\text{seeds} = [0, 1, 2].$$

Use the same DQN architecture, optimizer, replay buffer size, target update frequency, learning rate, discount factor, batch size, and number of episodes for both schedules. The only intended ablation variable is the exploration schedule.

A recommended exponential schedule is

$$\epsilon_t = \epsilon_{\min} + (\epsilon_{\max} - \epsilon_{\min}) \exp\left(-\frac{t}{\tau}\right),$$

where τ is small for fast decay and large for slow decay, and t denotes the training step.

Expected Plots and Tables

Your submitted report must include the following outputs from the python notebook and you must provide the explanation or analysis for the results.

- Random policy baseline mean and standard deviation.
- Reward learning curves for both exploration schedules and all seeds.
- Loss curves for both exploration schedules and all seeds.

- A table reporting final average return over the last 50 episodes for each schedule and seed.
- A summary table reporting mean and standard deviation across the three seeds for each schedule.

Discussion Questions

Answer the following questions in the final discussion section of the report. For detailed instructions, please refer to the notebook.

1. Which exploration schedule performed better on average across the three seeds?
2. How do the reward curves and loss curves differ? Why is loss alone not enough to decide whether the learned policy is good?
3. In MountainCar, why can moving away from the goal initially be useful?
4. Briefly state whether and how you used AI tools.
5. Sample-efficiency analysis. Sample efficiency is how effectively a model learns good policy from a limited amount of interaction data.

Submission

Submit the following:

- Completed `.ipynb` notebook.
- Completed `.pdf` or `.docx` report file.

Each submitted file must be named using the student ID of the person submitting it (e.g. 20269999.pdf). For team submissions, which may include up to two students, each student should submit the files using their own student ID as the file name.

The notebook should be executable from top to bottom after your TODOs are completed. If your results are noisy, explain the observed variance rather than hiding failed seeds.

AI-use Policy

You may use online documentation for setup issues. You may use AI tools for debugging, syntax help, or conceptual clarification, but you must not submit a fully AI-generated solution as your own. If you use AI tools, disclose what you used them for in the final discussion section. Undisclosed AI-generated code or analysis may be treated as an academic integrity issue.